



MSA University
Faculty of Engineering

ECE552 Mobile Communication Systems

Chapter 0 Course Outline

Prof. Fawzy Ibrahim
Dr. Abdelrahman El-Akhdar
Assoc. Prof. Nariman Abdelsalam

Electronics and Communication Department
October University for Modern Sciences and Arts (MSA)

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0.1 Course Description

ECE552 Mobile Communication Systems

Principles of wireless communication systems: basic concepts of cellular communications, cells, clusters, frequency reuse, multiplexing, cell sectorization, handoff, quality of service, number of the users served by a system, electromagnetic wave channel impairments, path loss models and interference in a cellular systems. Structures, main characteristics and evolutions of different generation of mobile communication systems: Second generation (2G) or GSM, Third generation (3G) or Universal Mobile Telecommunication System (UMTS), Fourth generation (4G) or Long-Term Evolution (LTE) and Fifth generation (5G) mobile network technologies.

Learning Outcomes (LOs):

After completing this course students will be able to:

LO1: Explain technical aspects, and operations.
LO2: Compare different generations of cellular mobile and personal communication technologies.
LO3: Differentiate among wireless communication systems and their standards.

Course prerequisites

ECE464 Antenna Theory and Design

0.2 Course Objectives

Upon a successful completion of this course, the student will be able to:

- Provide knowledge and understanding of the basic concepts of wireless telecommunications networks and their main components or modules.
- Comprehend the basics of mobile radio propagation environment and channel impairments then compute their effect on the mobile communication systems.
- Investigate the concepts of cellular networks, cell types and resources reuse such as frequency reuse systems then evaluate the system capacity..
- Differentiate between different cellular mobile generations: their structures, standardizations, different call or data Sequences or scenarios, the data evolution path, emerging technologies and services. These networks include:

1. The 2nd Generation (2G) or Global System for Mobile (GSM) communication technology basics and its evolution: High Speed Circuit Switched Data (HSCSD), General Packet Radio Service (GPRS) and Enhance Data rate for GSM Evolution (EDGE).
2. The 3rd Generation (3G) mobile technology or Universal Mobile Telecommunication System (UMTS) and its evolutions such as (HSDPA, HSUPA and HSPA+).
3. The 4th Generation (4G or LTE) mobile network with its advanced technologies such as: Carrier Aggregation (CA) and MIMO.
4. The 5th generation (5G) mobile with its emerging technologies: New multiple access technique (NOMA) and allocation of new spectrum (MMW) band.

0.3 Course Administration

Instructors: Assoc. Prof. Nariman Abdelsalam Room (D125)
Email: nbayoumy@msa.edu.eg
Lectures: Sundays and Mondays
Office Hrs: Sundays and Monday 8:00-11:00

Dr. Abdelrahman Mohamed El-Akhdar (D125)
Email: aelakhdar@msa.edu.eg
Office Hrs: Sundays and Mondays 8:00-12:00

TA: Eng. Hager Ahmed Room (G126)
hamorad@msa.edu.eg
Tutorials: Saturdays, Sundays, Mondays and Wednesdays
Office Hrs: Saturdays and Sundays 8:00 - 10:00
Text Books: 1. Martin Sauter, "From GSM to LTE-Advanced Pro and 5G", Fourth edition, 2024 John Wiley & Sons, Ltd.
2. Saad Z. Asif, "5G Mobile Communications, Concepts and Technologies ", Taylor & Francis Group, 2019.
3. T. Rappaport, "Wireless Communications: Principles and Practice", 2nd Edition, Prentice-Hall, 2002
Notes: Lecture slides and Assignments are on **E-Learning**.

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	1	**Chapter 0 Course Outline** 0.1 Course Description 0.2 Course Objectives 0.3 Course Administration 0.4 Course Outline 0.5 Grade Distribution **Chapter 1 Principles of wireless communications** 1.1 Wireless Communication Systems 1.1.1 Types of Wireless Communication Systems 1.1.2 Block Diagram of Typical Wireless digital com system 1.1.3 Transmission modes in wireless Communication 1. Simplex 2. Half-duplex 3. Full-duplex 1.1.4 Gain and Power Units 1.2 Cellular System Concepts 1.2.1 Geometries 1.2.2 Types of Mobile Cells 1.2.3 Frequency Reuse 1.2.4 Frequency Planning	***Problem Set #1*** Principles of wireless communications
	2	1.3 Number of Users in a Cellular System (System Capacity) 1.4 Handover or Handoff 1. with respect to the network 2. with respect to the link transfer 1.5 Channel Assignment Strategies (CAS)	

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	3	1.6 Interference in a Cellular System 1.6.1 Near End Far End Interference 1.6.2 Adjacent Channel Interference 1.6.3 Co-channel Interference 1.7 Improving Coverage & Capacity in Cellular Systems 1.Cell splitting 2. Cell sectoring 3. Zone microcell 4. Umbrella Cells 1.8 Repeaters and relay for Range Extension 1.9 Trunking and Grade of Service 1. Erlang B systems 2. Erlang C systems	***Problem Set #2*** Principles of wireless communications
	4	**Chapter 2 Principles of Mobile Communications** 2.1 Propagation Modeling in Cellular Systems 1.Line of Site 2. Ground wave propagation 3. Sky wave propagation 2.2 Electromagnetic Wave Channel Impairments 2.2.1 Attenuation or Free Space Loss 2.2.2 Large-scale fading (shadowing) 2.2.3 Small-scale fading (multipath) 2.2.4 Doppler Shift 2.3 Fading Mitigation Techniques 1. Diversity (time, freq., space) 2. Equalization	

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	5	2.4 Channel Propagation Models 2.4.1 Free Space Path Loss 2.4.2 Mobile Environment Model 2.4.3 Empirical Models 2.4.4 Statistical Models 2.5 Multiple Access Techniques 1. TDMA 2. FDMA 3. CDMA	***Problem Set #3*** Principles of wireless communications
	6	2.6 Mobile system network Architecture 2.6.1 Mobile Network Structure 2.6.2 Mobile Communication Site 2.6.3 Mobile Industry Companies 2.7 Mobile History or Generations 2.7.1First Generation (1G) 2.7.2Second Generation (2G) 2.7.3 Third Generation (3G) 2.7.4 Fourth Generation (4G) 2.7.5 Fifth Generation (5G)	

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	7	**Chapter 3 GSM** 3.1 GSM Specifications 3.1.1 The GSM Frequency Bands 3.1.2 Time Structure of the GSM System 3.1.3 GSM Transmission/Reception 3.1.4 The GSM power classes 3.2 GSM Network Architecture 3.2.1 Mobile Station (MS) 3.2.2 Base Station Subsystem (BSS) 3.2.3 Core Network (CN) 3.3 Mobile Call Sequence or Scenario 3.3.1 Mobile to Mobile 3.3.2 Land to Mobile 3.3.3 Mobile to Land	***Problem Set #4*** Principles of wireless communications
	8	3.4 GSM Network Location Areas 3.4.1 National GSM Service Area 3.4.2 PLMN Service Area 3.4.3 MSC/VLR Service Area 3.4.4 Location Areas (LAs) 3.4.5 BTS Service Area (Cell) 3.5 GSM Channel Types 3.5.1 Physical Channel 3.5.2 Logical Channels 3.6 GSM Evolution and Additional Features 3.6.1 (HSCSD) 2.9.2 (GPRS) 3.6.2 (EDGE)	

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	9	**Chapter 4** 4.1 3G / UMTS Specifications 4.1.1 3G Standardization 4.1.2 The 3G Frequency Bands 4.1.3 Multiple Access Technologies 4.1.4 Security Features 4.1.5Modulation Techniques 4.1.6Key Features and Enhancements 4.2 Wideband Code Division Multiple Access (WCDMA) 4.2.1 Spread Spectrum Principle 4.2.2 Multi-User DSSS (DS-CDMA) 4.2.3 Orthogonality Condition in DS-CDMA 4.3 WCDMA System Capacity 4.4 Codes in WCDMA 4.4.1 Channelization Codes 4.4 .2 Scrambling Codes 4.4.4 Code Reuse 4.4.3 Parameters of WCDMA	***Problem Set #5*** Principles of wireless communications
	10	4.5 UMTS System Architecture 4.5.1 UMTS High Level System Architecture 4.5.2 Basic UMTS Architecture 4.5.3 UMTS Interfaces 4.6 WCDMA Channels 4.6.1 Phys. Channels 4.6.2 log. Channels 4.6.3 Trans. channels 4.7 Power control in WCDMA 4.8 UMTS Evolution and Additional Features 4.8.1 (HSDPA) (R5-3.5G) 4.8.2 (HSUPA) (R6- 3.75G) 4.8.3 (HSPA or HSPA+) (R7 and R8 - 3.9G)	

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	11	**Chapter 5** 5.1 Quick Overview of Wireless Standards 5.2 4G / LTE Specifications 5.2.1 4G Standardization 5.2.2 The 4G Frequency Bands 5.2.3 Multiple Access Technologies 5.2.4 Security Features 5.2.5 Modulation Techniques5.2.6 Key Features and Enhancements 5.3 LTE System Architecture 5.3.1 Evolved UTRAN (E-UTRAN) 5.3.2 Evolved Packet Core (EPC) 5.4 Multi-Carrier and OFDM Transmission 5.4.1 Single Carrier Transmission 5.4.2 Multi-Carrier Transmission 5.4.3 Orthogonal Frequency Division Multiplexing (OFDM) 5.4.4 Guard time - Cyclic Prefix (CP) 5.4.5 Block diagram of full OFDM System 5.4.6 Difference between OFDM and OFDMA 5.5 LTE Downlink and Uplink Transmission 5.5.1 LTE downlink Transmission □ OFDMA 5.5.2 LTE Uplink Transmission □ SC-FDMA 5.6 LTE Resource Grid and Frame Structure 5.6.1 Time–Frequency Representation 5.6.2 LTE Frame Structure 5.6.3 LTE Bandwidth Allocation Options 5.6.4 LTE UE categories (downlink and uplink) 5.6.5 LTE Key Parameters	***Problem Set #6*** Principles of wireless communications

0.4 Course Outline

Wk	Lec	Items/Topics	Assignments
	12	**Chapter 6** 6.1 Drivers for 5G 6.2 5G Roadmap 6.3 5G Requirements by IMT-2020 6.4 5G Key Enabling Technologies 6.4.1 Evolution of Existing RATs 6.4.2 Cloud Radio Access Network (C-RAN) 6.4.3 Non-Orthogonal Multiple Access (NOMA) 6.4.4 Evolution Hyperdense Small-Cell Deployment 6.4.5 Self Organizing Networks (SON) 6.4.6 Device-to-Device (D2D) Communication 6.4.7 Developing Millimeter-Wave RATs 6.4.8 Machine Type Communication (MTC) 6.4.9 Redesigning Backhaul Links 6.4.10 Allocation of New Spectrum for 5G 6.5 Non-Orthogonal Multiple Access (NOMA) 6.5.1 Basics of NOMA 6.5.2 SC and SIC 6.5.3 NOMA Scheme 6.5.4 NOMA for Downlink 6.5.5 NOMA for Uplink 6.5.6 Orthogonal Multiple Access (OMA) vs. NOMA 6.5.7 A Numerical Example of NOMA 6.5.8 Advantages of NOMA 6.5.9NOMA Combined with MIMO	***Problem Set #6*** Principles of wireless communications

05 Possible Term Project Researches

Selected Tonics for the Term Project

1.Hard vs. Soft Handover in 2G/3G.
2.Handover in LTE (X2/S1 handover).
3.5G Mobility Management and Beam Switching.
4.Handover Performance Metrics (Delay, Dropped Calls).
5.WCDMA Spreading and Rake Receivers.
6.HSPA Enhancements.
7.QoS Classes in UMTS.
8.Applications: Mobile Internet & Video Calling.
9.MIMO technology in 4G.
10.Carrier aggregation in LTE-Advanced.
11.Voice over LTE (VoLTE) technology.
12.OFDM Systems.
13.5G network architecture and core network (5GC).
14.5G frequency bands (FR1 vs FR2, mmWave).
15.Beam forming and massive MIMO in 5G.
16.Network slicing in 5G.
17.5G use cases: eMBB, mMTC, URLLC.
18.Comparison of Latency, Data Rates, and Capacity (2G–5G)
19.Energy efficiency and sustainability in mobile networks.
20.Any advanced topics related to the mobile and wireless systems.

0.5 Grade Distribution

Evaluation	Written	Quizzes	Assignments and Participation	Term Project	Total
Midterm	20	10	5	-	35
Final	40	10	5	10	65
Total Score	60	20	10	10	100

Important university regulation:

- **Absence from lectures and/or tutorials shall not exceed 25%.**
- **Students who exceed the 25% limit shall not be allowed to take the final examination.**